

Weather & Air Quality Station — Cheat Sheet

Chain K — Hardware & Systems Foundations · one-page recall, not first-time learning

Sensors

Fact	Detail
Rain gauge measurement	Raw tips × mm-per-tip calibration = real rainfall depth
BME280 buying red flag	No-datasheet clones drift badly over time

Solar Power & Battery Sizing

Formula	Meaning
Daily consumption	Average current (mA) × 24h
Daily harvest	Panel watts × real sun-hours × efficiency, converted to mAh
Net balance	Harvest – consumption — must stay positive through the CLOUDIEST week
Autonomy	(Battery mAh × usable fraction) ÷ daily deficit

Connectivity

Option	Tradeoff
Wifi	Higher bandwidth, easy integration, limited range, more power
LoRa	Much longer range, lower power, far lower bandwidth — fits small/rare data

Weatherproofing & Reliability

Detail	Reason
BME280 vent membrane	A sealed box reads barometric pressure wrong
Condensation fix	Same vent membrane that fixes the pressure problem
Waterproofing pattern	Reused directly from Walkie-Talkie-Build (IP67 + sealed glands)

Chain K · Weather-And-Air-Quality-Station · full reasoning in LESSON.pdf